Homework 9: Random Variables

STAT 141, Week 11

Your name here

Assigned: Thursday, November 12th

Due: Thursday, November 19th at 9:00am on Gradescope as a PDF

In this homework assignment you will:

- Calculate the expectation, variance, and standard deviation of random variables
- Calculate probabilities and quantiles for Normal distributions in `R`
- Calculate and interpret z-scores
- Use visualizations to assess Normality

Add any collaborators here!

I collaborated with…

Remember that you are not allowed to use any internet resources or generative AI models (like ChatGPT or Claude) on your assignment. While there are many ways to achieve tasks in R, you may only use packages and functions that have been covered in this course. If you are unsure of how to approach an exercise or of what function to use, help is available to you! Please visit office hours, collaborate with your peers, or post in the Slack channel.

Exercises

Exercise 1: Discrete Random Variables

Supposed X is a discrete random variable with the following distribution table:

X	$P(X = x)$
-3	0.2
3	0.5
10	0.3

a. What is $E(X)$? [Only text/math is needed for this question]

b. What are $Var(X)$ and $SD(X)$? [Only text/math is needed for this question]

c. Suppose $Y = X^2$. What is the distribution table for Y? [Only text/math is needed for this question]

Y	$P(Y = y)$

d. What is $E(Y) = E(X^2)$? [Only text/math is needed for this question]

e. Using your answers for parts (a) and (d), calculate $E(X^2) - [E(X)]^2$. Compare your answer to what you got for $Var(X)$ in part (b). [Only text/math is needed for this question]

Exercise 2: Normal Distribution Warm Up

The average daily high temperature in June in Los Angeles (LA) is 77 degrees Fahrenheit, with a standard deviation of 5 degrees. Suppose that the temperatures in June closely follow a normal distribution, i.e., $F \sim \text{Normal}(\mu = 77, \sigma = 5)$.

a. What is the probability of observing an 83 degree or higher day in LA during a randomly chosen day in June? (*Hint: remember that the R functions `pnorm()`, `dnorm()`, and `qnorm()` can help you calculate quantities related to Normal distributions.*) [Code + text to state your answer clearly is needed for this question]

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b. How cool are the coldest 10% of the days (days with lowest high temperature) during June in LA? (*Hint: remember that the R functions `pnorm()`, `dnorm()`, and `qnorm()` can help you calculate quantities related to Normal distributions.*) [Code + text to state your answer clearly is needed for this question]

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Exercise 3: Computing Inverse Probabilities

The Graduate Record Exam (GRE) is a standardized test which serves as an admission requirement for many graduate schools, and has an analytic writing section and two multiple choice sections: Verbal Reasoning and Quantitative Reasoning. Due to forced Normalization in scores, both sections are approximately Normal. In a given year, suppose the mean score for Verbal Reasoning section was 462 with a standard deviation of 119, and the mean score for the Quantitative Reasoning was 584 with a standard deviation of 151.

We will use the above information to continue exploring the use of the R functions `pnorm` and `qnorm`. As a reminder:

- `pnorm(q)` will return the **cumulative probability** $P[X \leq q]$ for $X \sim \text{Normal}(0, 1)$, i.e. the cumulative area under the normal curve to the *left* of a given point on the x-axis, q.
- `qnorm(p)` will return the **quantile** (on the x-axis) of a given **cumulative probability** p. I.e. it will find the point q where $p * 100$ percent of the area under the curve is to the left of q.

By default, the curve considered is the *standard* normal distribution but we can change the center and standard deviation of the normal distribution considered by adding `mean` and `sd` parameters.

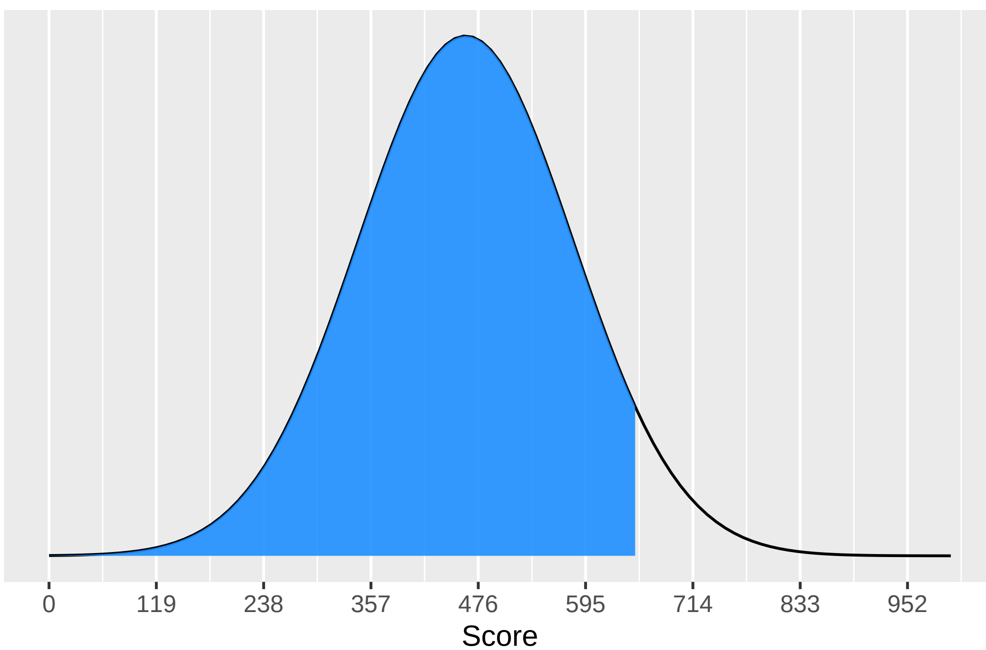
Using `pnorm`

For instance, say a student earns a score of 650 on the Verbal Reasoning section of the exam. We often describe the cumulative probability to the *left* of this score as the student's *percentile*. We can calculate this using `pnorm()`:

```
pnorm(650, mean = 462, sd = 119)
```

```
[1] 0.9429273
```

We see that this student is in the 94th percentile. That is, 94% of the values in the distribution of verbal reasoning scores are to the left of 650. This is shown as the shaded region in the visualization below.

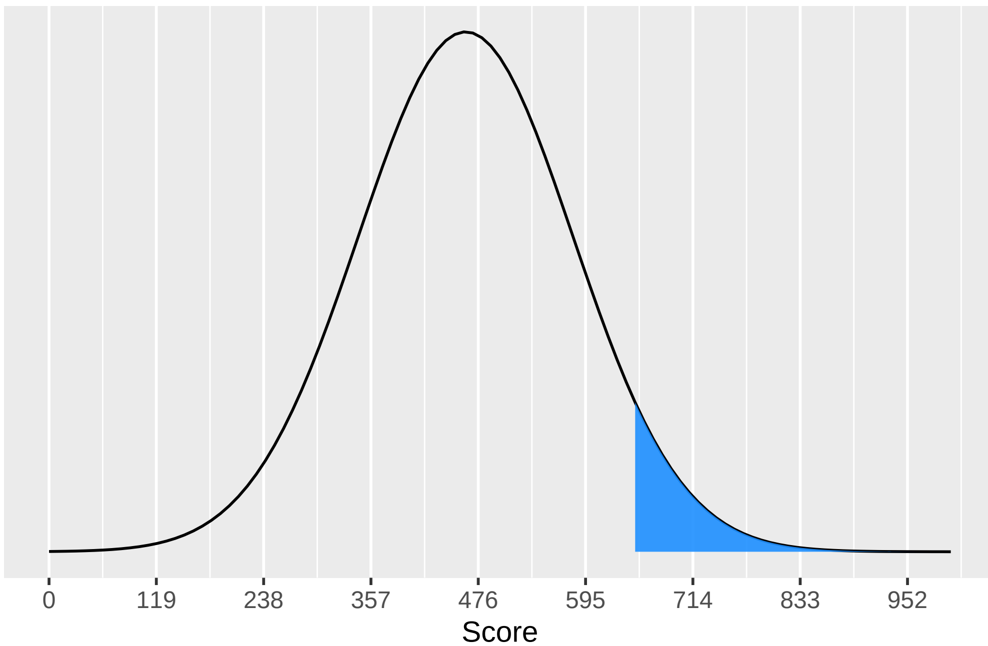


If we wanted to know what percentage of students scored **better** than 650, we calculate 1 minus the percentage to the left of the value.

```r
1 - pnorm(650, mean = 462, sd = 119)
```

```
[1] 0.05707268
```

The visualization below shows that this is, appropriately, in the right tail of the distribution.



Using `qnorm`

Now lets imagine that a student is told that they scored in the 57% percentile on the Quantitative Reasoning section of the exam. We can use **`qnorm`** to do the reverse of our previous task: find the score (the "inverse probability") for this student given their "percentile" value.

```r
qnorm(0.57, mean = 584, sd = 151)
```

```
[1] 610.6325
```

This student's score is thus a 610.

a. What is the percentage of the students who score between 590 and 630 on the Quantitative Reasoning section? (*Hint: draw a picture of the area under the curve corresponding to this quantity, and consider how we can calculate it using either `pnorm` or `qnorm`.*) [Only code required]

b. What is the probability that a randomly selected student earns obtains a score greater than 600 on both multiple choice sections? **Assume (somewhat unrealistically!), that scores on the two sections are independent**. [Only code required]

c. What score is needed on the Verbal Reasoning section to place in the **top** 10%? What score is needed to place in the top 10% on the Quantitative Reasoning section? [*Code + brief written response to distinguish which value is which*]

Exercise 4: Standard Normal

Often it is useful to work with the *Standard Normal*. This is a normal distribution with a mean of 0 and a standard deviation of 1. If we know that a value x is drawn from a normal distribution with a mean of μ and a standard deviation of σ, we can standardize it using the following formula:

$$z = \frac{x - \mu}{\sigma}$$

These standardized values are called **z-scores**. We can calculate the z-score for the student who scored a 650 on the Verbal Reasoning section of the exam.

```
score = 650
mu = 462
sigma = 119

(score - mu) / sigma
```

```
[1] 1.579832
```

Note that for scores that are less than the mean, the z-score will be negative. Imagine a student scored a 550 on the Quantitative Reasoning section.

```
score = 550
mu = 584
sigma = 151

(score - mu) / sigma
```

```
[1] -0.2251656
```

They have a z-score of -0.23.

a. A college senior scores 620 on the Verbal Reasoning section and 670 on the Quantitative Reasoning section. Calculate the student's z-scores for each section. [*Code + brief written response to distinguish which value is which*]

b. In general, how should we interpret z-scores? (*Hint: think about why normalizing the standard deviation to 1 and mean to 0 is useful*). On which section did the student in the previous part do better on (relative to the student's peers)? [*Only written response is required*]

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c. Find the student's score percentiles for the two sections. You can use either their raw scores or their z-scores. [*Code + brief written response to distinguish which value is which*]

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Exercise 5: Checking Normality in Fast Food Data

For the remainder of this assignment we'll be working with fast food data. This data set contains data on 515 menu items from some of the most popular fast food restaurants worldwide. Let's take a quick peek at the first few rows

```r
data(fastfood)
glimpse(fastfood)
```

```
Rows: 515
Columns: 17
$ restaurant  <chr> "Mcdonalds", "Mcdonalds", "Mcdonalds", "Mcdonalds", "Mcdon~
$ item        <chr> "Artisan Grilled Chicken Sandwich", "Single Bacon Smokehou~
$ calories    <dbl> 380, 840, 1130, 750, 920, 540, 300, 510, 430, 770, 380, 62~
$ cal_fat     <dbl> 60, 410, 600, 280, 410, 250, 100, 210, 190, 400, 170, 300,~
$ total_fat   <dbl> 7, 45, 67, 31, 45, 28, 12, 24, 21, 45, 18, 34, 20, 34, 8, ~
$ sat_fat     <dbl> 2.0, 17.0, 27.0, 10.0, 12.0, 10.0, 5.0, 4.0, 11.0, 21.0, 4~
$ trans_fat   <dbl> 0.0, 1.5, 3.0, 0.5, 0.5, 1.0, 0.5, 0.0, 1.0, 2.5, 0.0, 1.5~
$ cholesterol <dbl> 95, 130, 220, 155, 120, 80, 40, 65, 85, 175, 40, 95, 125, ~
$ sodium      <dbl> 1110, 1580, 1920, 1940, 1980, 950, 680, 1040, 1040, 1290, ~
$ total_carb  <dbl> 44, 62, 63, 62, 81, 46, 33, 49, 35, 42, 38, 48, 48, 67, 31~
$ fiber       <dbl> 3, 2, 3, 2, 4, 3, 2, 3, 2, 3, 2, 3, 3, 5, 2, 2, 3, 3, 5, 2~
$ sugar       <dbl> 11, 18, 18, 18, 18, 9, 7, 6, 7, 10, 5, 11, 11, 11, 6, 3, 1~
$ protein     <dbl> 37, 46, 70, 55, 46, 25, 15, 25, 25, 51, 15, 32, 42, 33, 13~
$ vit_a       <dbl> 4, 6, 10, 6, 6, 10, 10, 0, 20, 20, 2, 10, 10, 10, 2, 4, 6,~
```

```
$ vit_c       <dbl> 20, 20, 20, 25, 20, 2, 2, 4, 4, 6, 0, 10, 20, 15, 2, 6, 15~
$ calcium     <dbl> 20, 20, 50, 20, 20, 15, 10, 2, 15, 20, 15, 35, 35, 35, 4, ~
$ salad       <chr> "Other", "Other", "Other", "Other", "Other", "Other", "Oth~
```

You'll see that for every observation there are 17 measurements, many of which are nutritional facts. We'll focus on just three columns to get started: restaurant, calories, calories from fat. And let's focus on just products from McDonalds and Dairy Queen.

```
mcdonalds <- fastfood %>%            # Data frame with just McDonald's data
  filter(restaurant == "Mcdonalds")
dairy_queen <- fastfood %>%          # Data frame with just Dairy Queen data
  filter(restaurant == "Dairy Queen")
```

a. Make plots to visualize the *distribution* of the amount of *calories from fat* among the menu items at McDonald's and Dairy Queen (one plot per restaurant). **How do their centers, shapes, and spreads compare?** [*Code + written response required*]

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b. In your description of the distributions, did you use words like *bell-shaped* or *normal*? It's tempting to say so when faced with a unimodal symmetric distribution. To see how accurate that description is, we can plot a normal distribution curve on top of a histogram to see how closely the data follow a normal distribution.

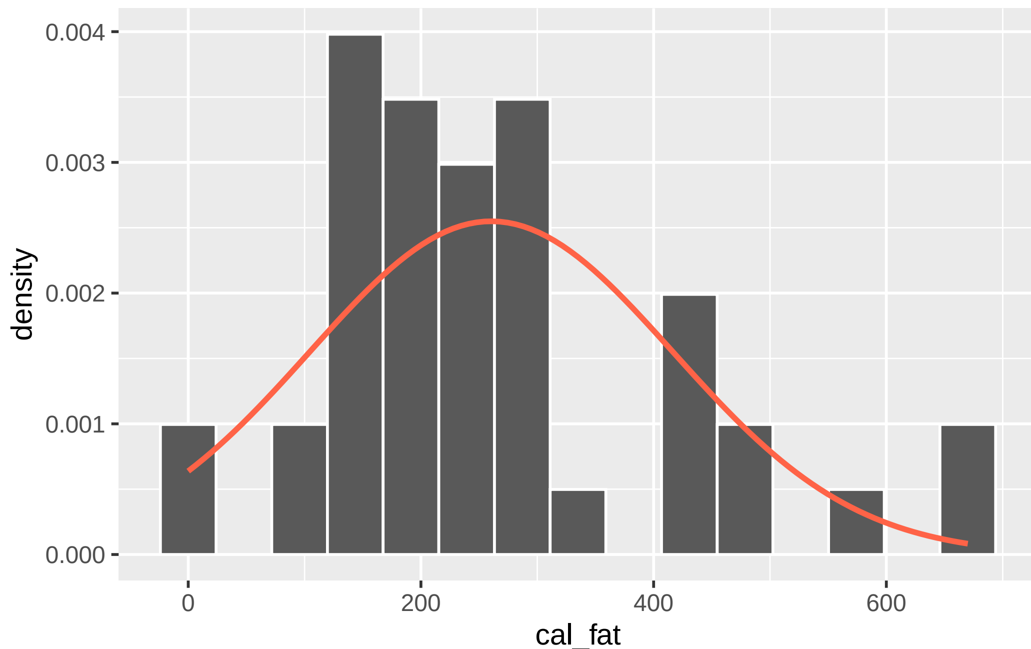
This normal curve should have the same mean and standard deviation as the data. We'll focus on calories from fat from Dairy Queen products, so let's store them as a separate object and then calculate some statistics that will be referenced later.

```
dq_mean <- mean(dairy_queen$cal_fat)
dq_sd <- sd(dairy_queen$cal_fat)
```

Next, we'll make a *density histogram* with our empirical data to use as the backdrop, and use `stat_function` to overlay a theoretical normal probability curve. The difference between a frequency histogram and a density histogram is that while in a frequency histogram the *heights* of the bars add up to the total number of observations, in a density histogram the *areas* of the bars add up to 1. This allows us to properly overlay a normal distribution curve over the histogram, since the curve is a normal probability density function that also has area under the curve of 1.

Frequency and density histograms both display the same exact shape; they only differ in their y-axis. You can verify this by comparing the frequency histogram you constructed earlier and the density histogram created by the commands below.

```
ggplot(data = dairy_queen, aes(x = cal_fat)) +
  geom_histogram(aes(y = ..density..),  # Tells ggplot to make a *density* histogram
                 color = "white", bins = 15) +
  stat_function(fun = dnorm,  # Tells ggplot: take the `dnorm()` function and plot it...
                args = c(mean = dq_mean, sd = dq_sd), # ... with this mean and sd
                color = "tomato", lwd = 1)
```



The first layer of the plot above is a density histogram: the data from Dairy Queen that we observe. The second layer is a statistical function: the theoretical density of the normal curve, `dnorm`. We specify that we want the curve to have the same mean and standard deviation as the column of calories from fat. You can think of this as us finding the theoretical Normal distribution that *best fits* our data according to its mean and standard deviation.

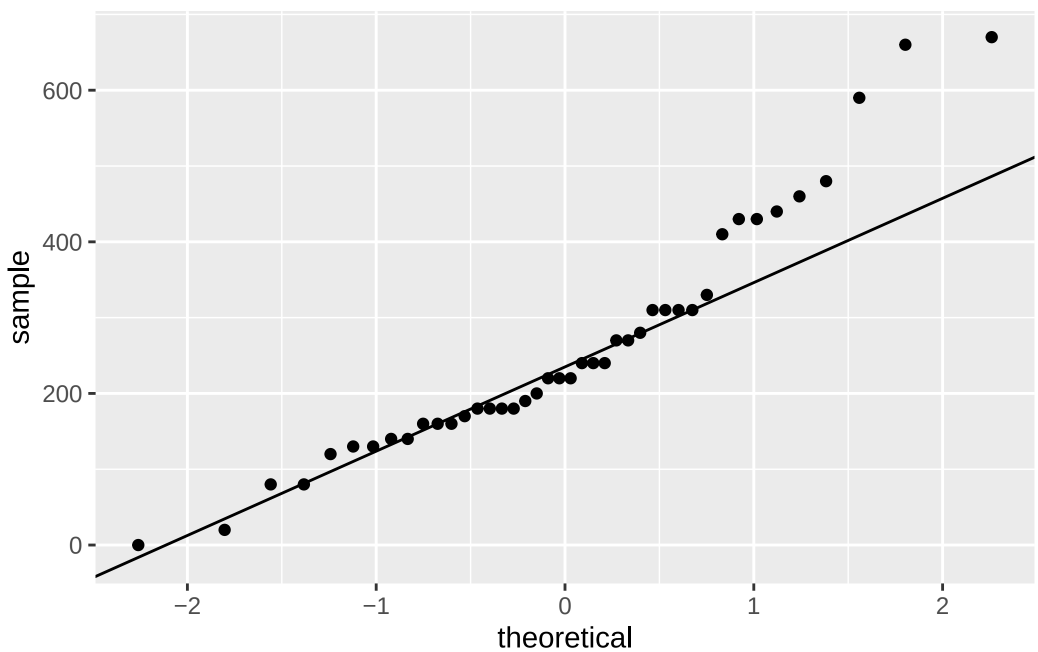
Based on this plot, does it appear that the data follow a nearly normal distribution? **If you are unsure, describe which aspects may suggest normality, and which do not.** [*Only written response required*]

Exercise 6: Checking Normality with Q-Q Plots

Eyeballing the shape of the histogram is one way to determine if the data appear to be nearly normally distributed, but it can be frustrating to decide just how close the histogram is to the curve. It's especially tricky when we don't have a very large sample - there are only 42 values in the histogram above.

We've already learned an alternative approach: using a Q-Q plot!

```
ggplot(data = dairy_queen, aes(sample = cal_fat)) +
  geom_qq() +
  geom_qq_line() +
  labs(x = "theoretical", y = "sample")
```



We used this as a black box tool to evaluate the "Normality" assumption for linear models. Now we have the tools to understand what is going on under the hood. Q-Q stands for "quantile-quantile". The x-axis values correspond to the *quantiles of a theoretically normal curve with mean 0 and standard deviation 1* (i.e., the standard normal distribution). The y-axis values correspond to the *quantiles of the unstandardized sample data we're providing* (however, even if we were to standardize the sample data values, the Q-Q plot would look identical).

A data set that is nearly normal will result in a probability plot where the points closely follow a diagonal line. Any deviations from normality leads to deviations of these points from that line.

Above, the plot for Dairy Queen's calories from fat shows points that tend to follow the line but with some errant points towards the upper tail. We're left with the same problem that we encountered with the histogram above: **how close is close enough**?

A useful way to address this question is to rephrase it as: what do probability plots look like for data that I *know* came from a normal distribution? We can answer this by simulating data from a normal distribution using `rnorm`.

```
set.seed(123)
sim_norm <- rnorm(n = nrow(dairy_queen), mean = dq_mean, sd = dq_sd)
sim_norm_df <- data.frame(sim_norm)
```

The first argument of `rnorm()` indicates how many numbers you'd like to generate, which we specify to be the same number of menu items in the `dairy_queen` data set using the `nrow()` function. The last two arguments determine the mean and standard deviation of the normal distribution from which the simulated sample will be generated.

a. Make a Q-Q plot of `sim_norm`. What should we expect to see? Why do all of the points not fall perfectly along the line, even though this data is drawn from a true Normal distribution? [*Code + written response required*]

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b. How does the plot from part (a) compare to the Q-Q plot for the real data? [*Only written response required*]

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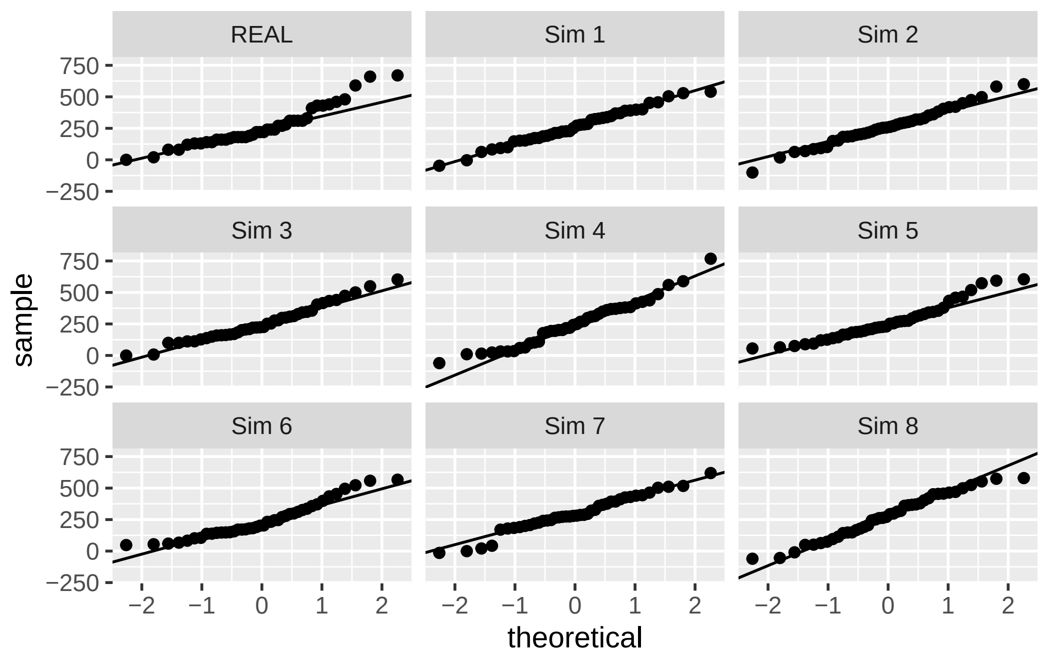
c. Even better than comparing the original plot to a single plot generated from a normal distribution is to compare it to plots from several samples. The code below shows the Q-Q plot corresponding to the original Dairy Queen data in the top left corner, and the Q-Q plots of 8 different simulated normal data samples.

```r
set.seed(123)
sim_data8 <- data.frame(value = rnorm(n = 8*nrow(dairy_queen), mean = dq_mean,
                                      sd = dq_sd),
                        sample = paste("Sim", rep(1:8, each = nrow(dairy_queen))))
comp_data <- rbind(data.frame(value =  dairy_queen$cal_fat,
                              sample = "REAL"),
                   sim_data8)
ggplot(comp_data, aes(sample = value)) +
  geom_qq() +
  geom_qq_line() +
  labs(x = "theoretical", y = "sample") +
  facet_wrap(~sample)
```



Does the Q-Q plot for the calories from fat stand out from the plots created with simulated Normal data? That is, do the plots provide evidence that the calories from fat are nearly normal?

[Only written response required]

Exercise 7: Normal Probabilities

Now we have a collection of tools to judge whether or not a variable is normally distributed.

Why should we care?

Statisticians know a lot about the normal distribution! Once you decide that a random variable is approximately normal, you can readily answer all sorts of questions about that variable related to probability. For example, consider the question *"What is the probability that a randomly chosen Dairy Queen product has more than 600 calories from fat?"*

If we assume that the calories from fat from Dairy Queen's menu are normally distributed (a very close approximation is also okay), we can find this probability by calculating a probability in the Normal distribution. In R, this is done in one step with the function `pnorm()`.

```
1 - pnorm(q = 600, mean = dq_mean, sd = dq_sd)
```

```
[1] 0.01501523
```

Note that the function `pnorm()` gives the area under the normal curve *less than a given value*, `q`, for a normal distribution with a given mean and standard deviation. Since we're interested in the probability that a Dairy Queen item has *more than* 600 calories from fat, we have to take one minus that probability.

Assuming a normal distribution has allowed us to calculate a *theoretical probability*, based on the theoretical normal distribution that we claim our data comes from. If we want to calculate the probability *empirically*, we simply need to determine how many observations fall above 600 then divide this number by the total sample size.

```
dairy_queen %>%
  filter(cal_fat > 600) %>%
  summarise(percent = n() / nrow(dairy_queen))
```

```
# A tibble: 1 x 1
  percent
    <dbl>
1  0.0476
```

Although the probabilities are not exactly the same, they aren't terribly far apart. The closer that your distribution is to being normal, the more accurate the theoretical probabilities will be.

Using both the empirical approach and the theoretical approach, calculate the probability that a randomly chosen item from **McDonalds** will have more than 1000 calories (measured in the `calories` column). Do the same to calculate the probability that a randomly chosen item from Dairy Queen will have more than 1000 calories. (*four calculations in all*). **Which restaurant had a closer agreement between the two methods?**

[Code + written response required]